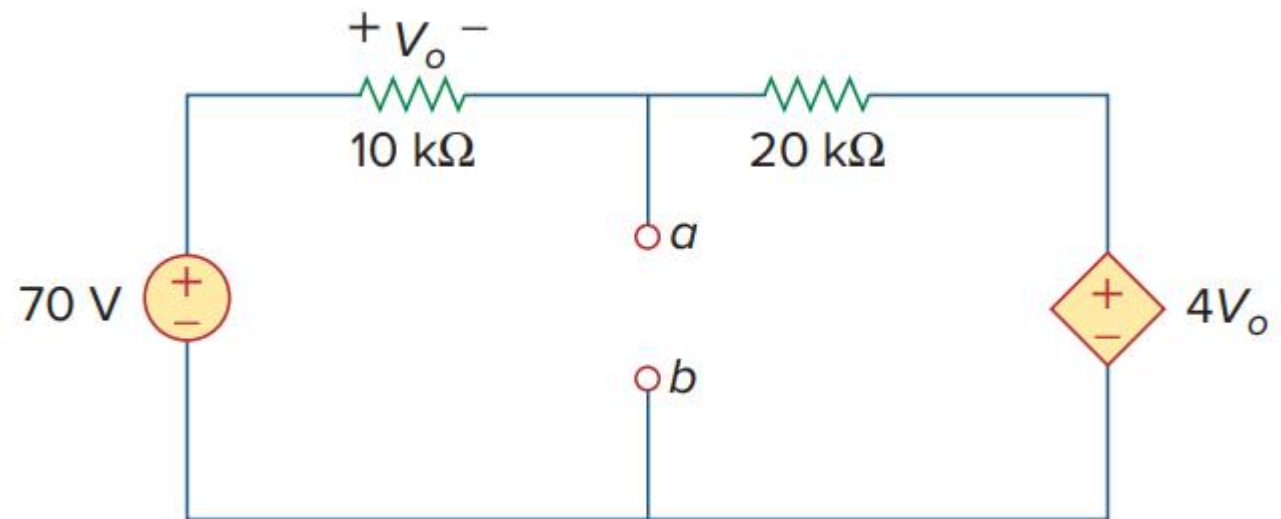
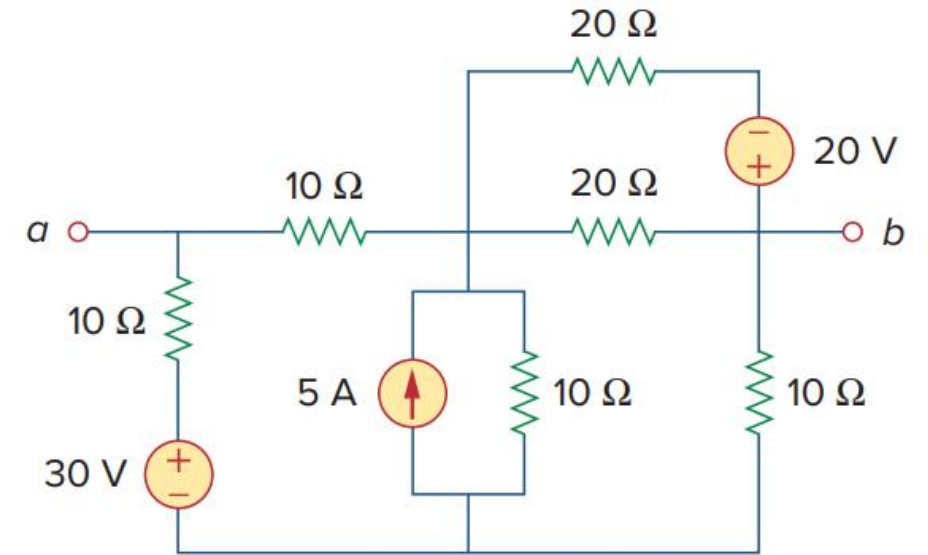
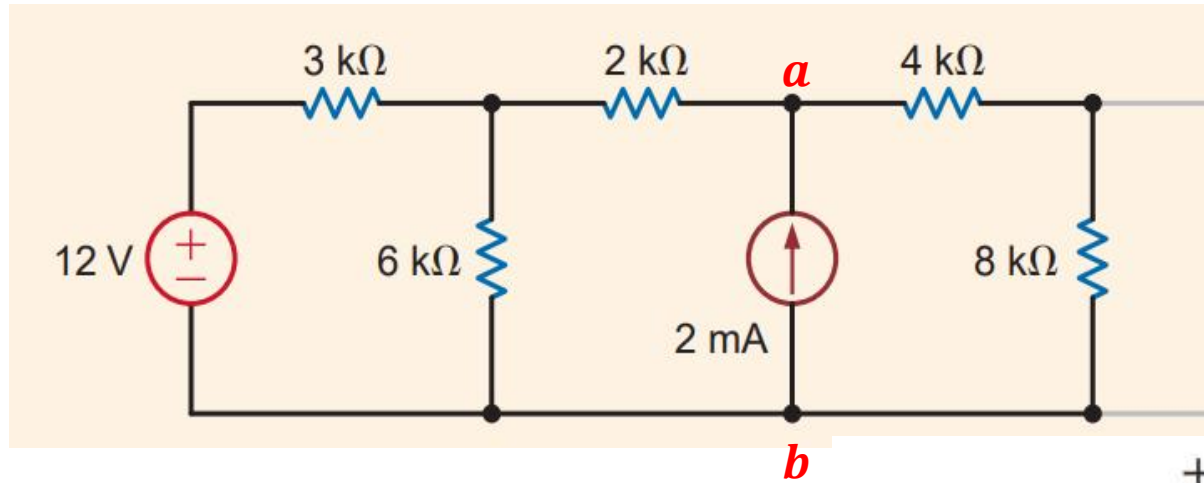
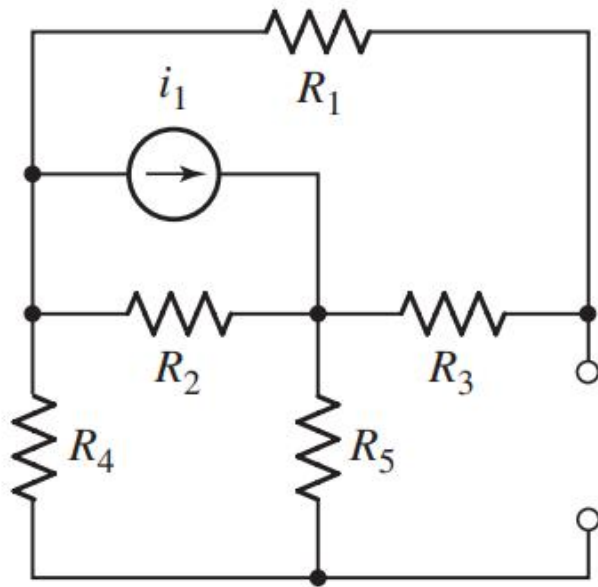


HW 9

Find V_{TH} and R_{TH} between **a** and **b** in the following circuit

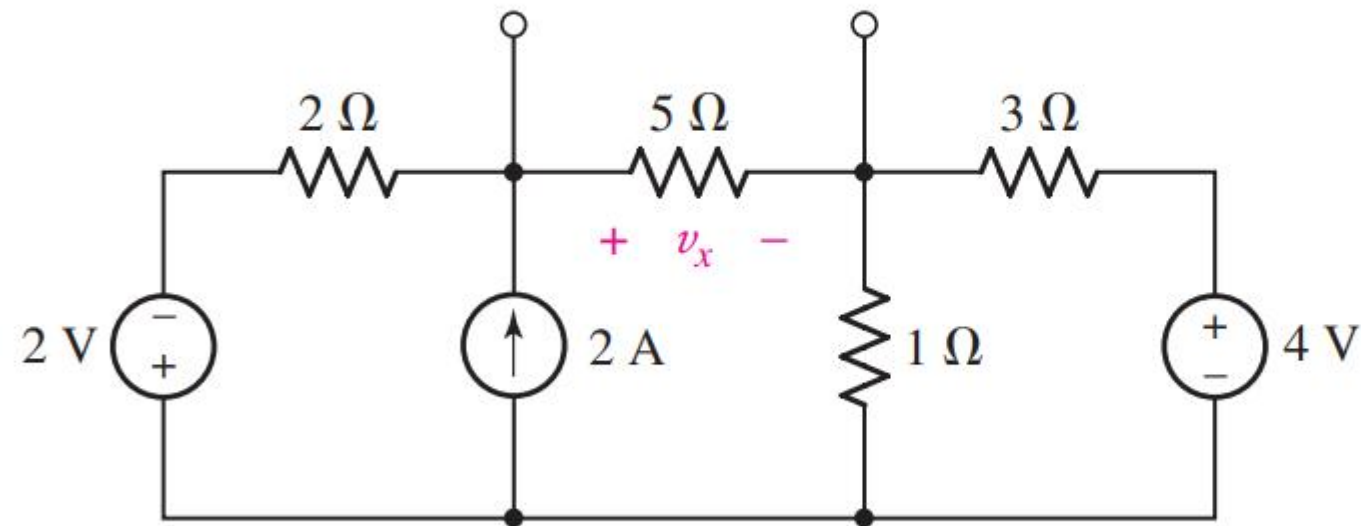


1. (a) Employ Thévenin's theorem to obtain a two-component equivalent for the network shown in Fig. 5.73. (b) Determine the power supplied to a $1\text{ M}\Omega$ resistor connected to the network if $i_1 = 19\text{ }\mu\text{A}$, $R_1 = R_2 = 1.6\text{ M}\Omega$, $R_2 = 3\text{ M}\Omega$, and $R_4 = R_5 = 1.2\text{ M}\Omega$. (c) Verify your solution by simulating both circuits with PSpice or another appropriate CAD tool.



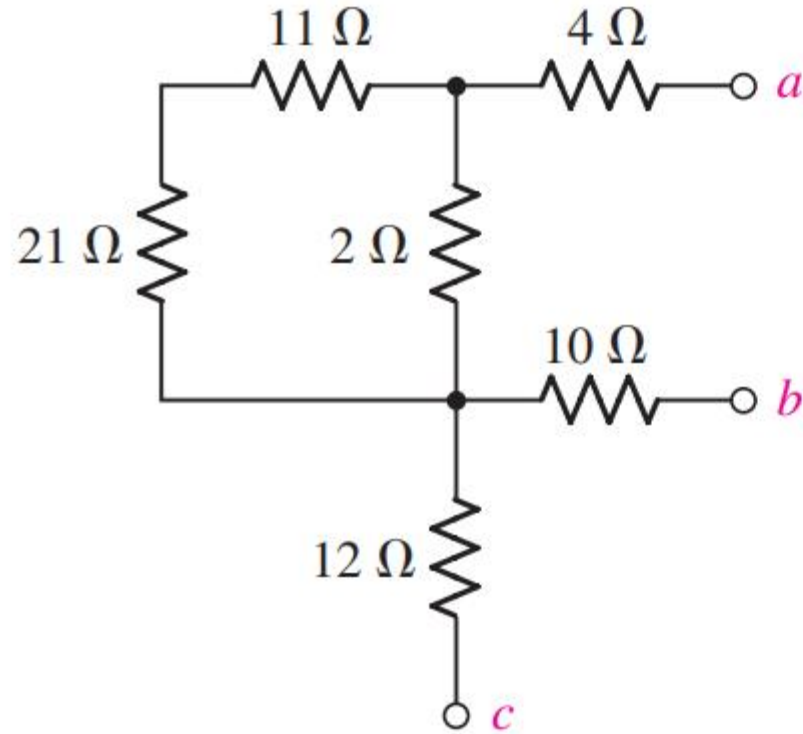
■ FIGURE 5.73

32. Determine the Thévenin equivalent of the network shown in Fig. 5.74 as seen looking into the two open terminals.

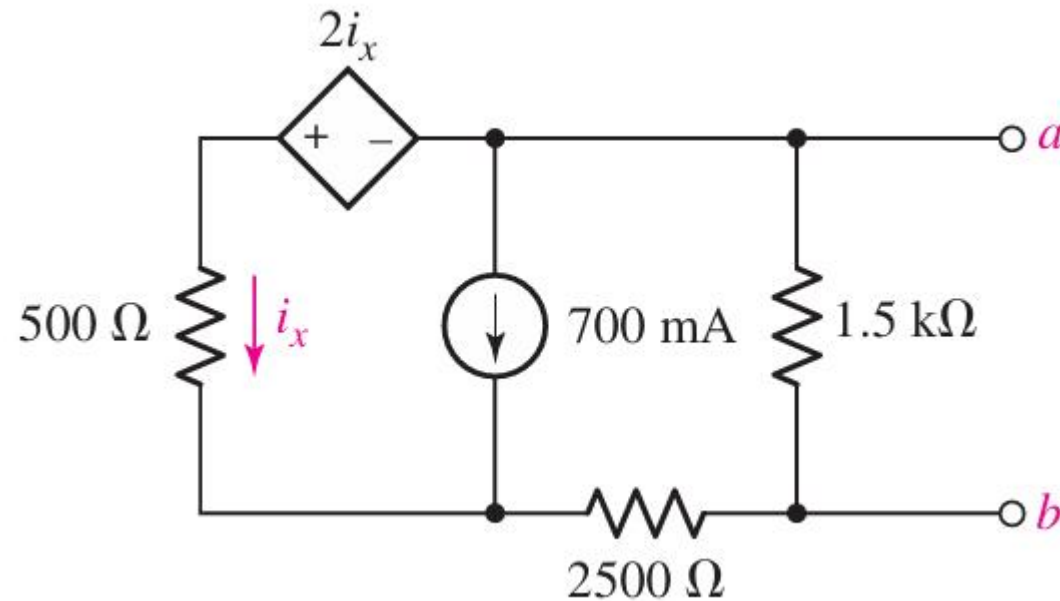


■ FIGURE 5.74

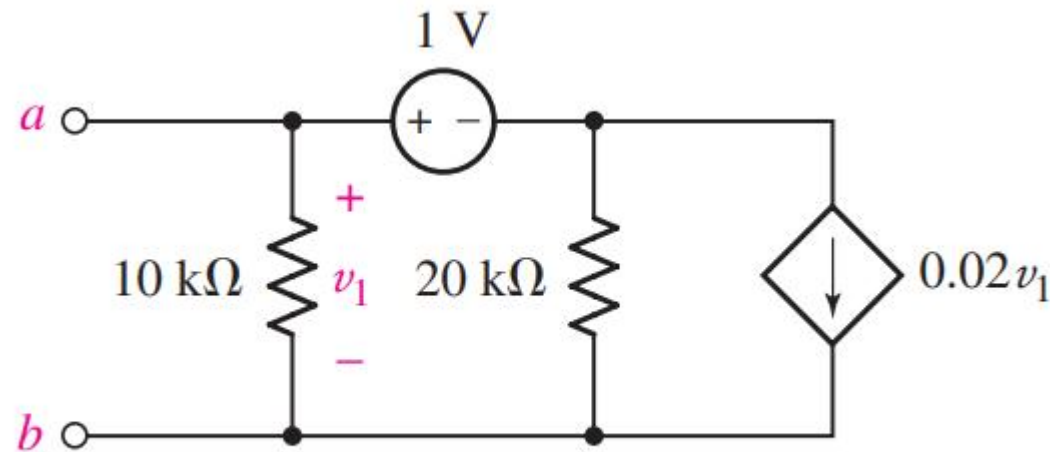
38. With regard to the network depicted in Fig. 5.79, determine the Thévenin equivalent as seen by an element connected to terminals (a) a and b ; (b) a and c ; (c) b and c . (d) Verify your answers using PSpice or other suitable CAD tool. (Hint: Connect a test source to the terminals of interest.)



40. Determine the Norton equivalent of the circuit drawn in Fig. 5.81 as seen by terminals a and b . (There should be no dependent sources in your answer.)

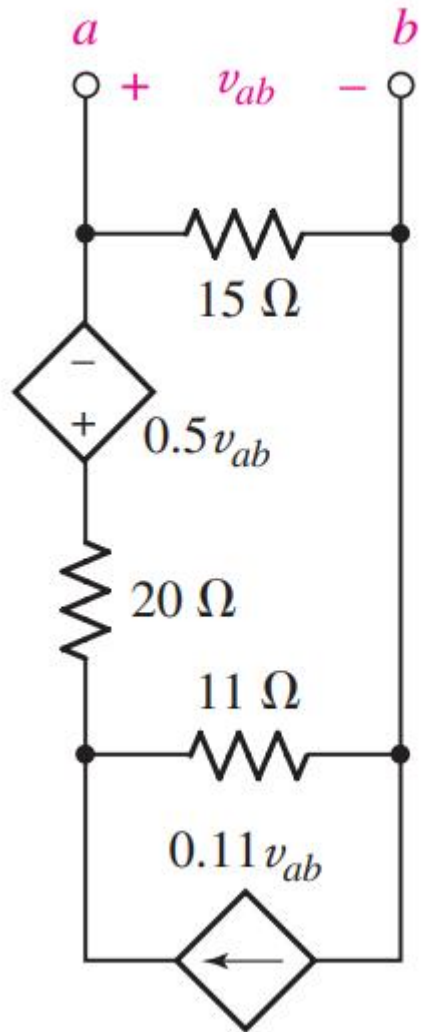


41. With regard to the circuit of Fig. 5.82, determine the power dissipated by (a) a $1\text{ k}\Omega$ resistor connected between a and b ; (b) a $4.7\text{ k}\Omega$ resistor connected between a and b ; (c) a $10.54\text{ k}\Omega$ resistor connected between a and b .

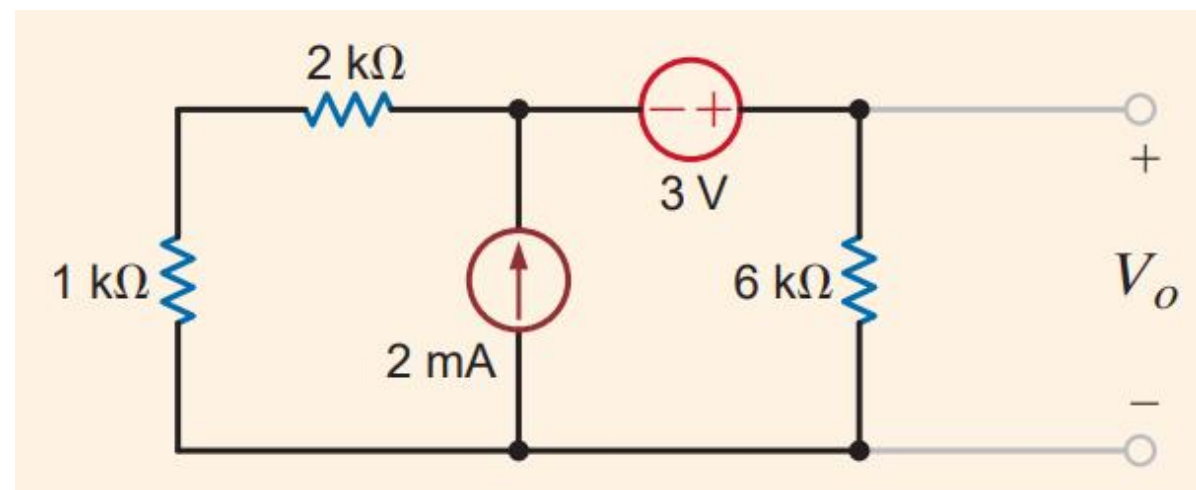
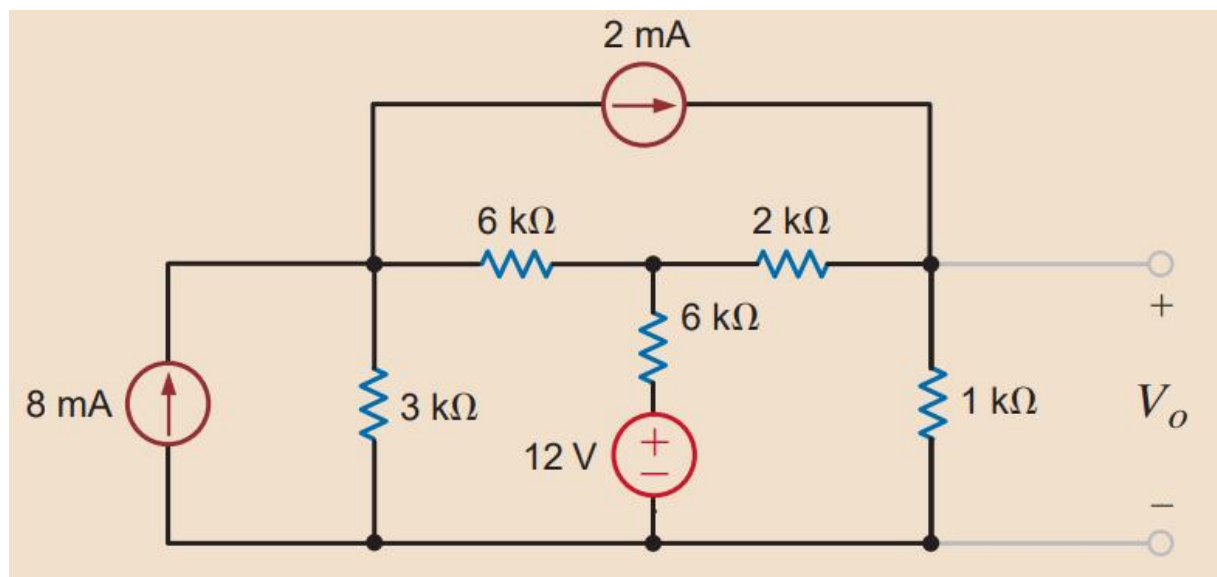


■ **FIGURE 5.82**

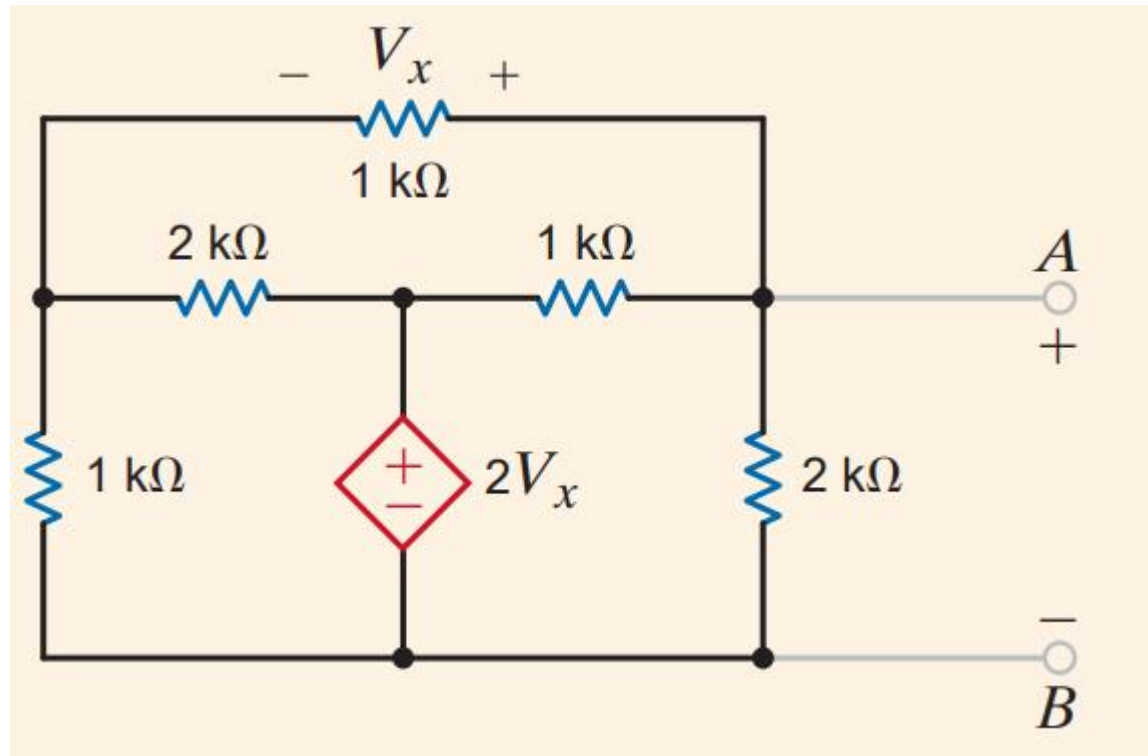
42. Determine the Thévenin and Norton equivalents of the circuit shown in Fig. 5.83, as seen by an unspecified element connected between terminals a and b .



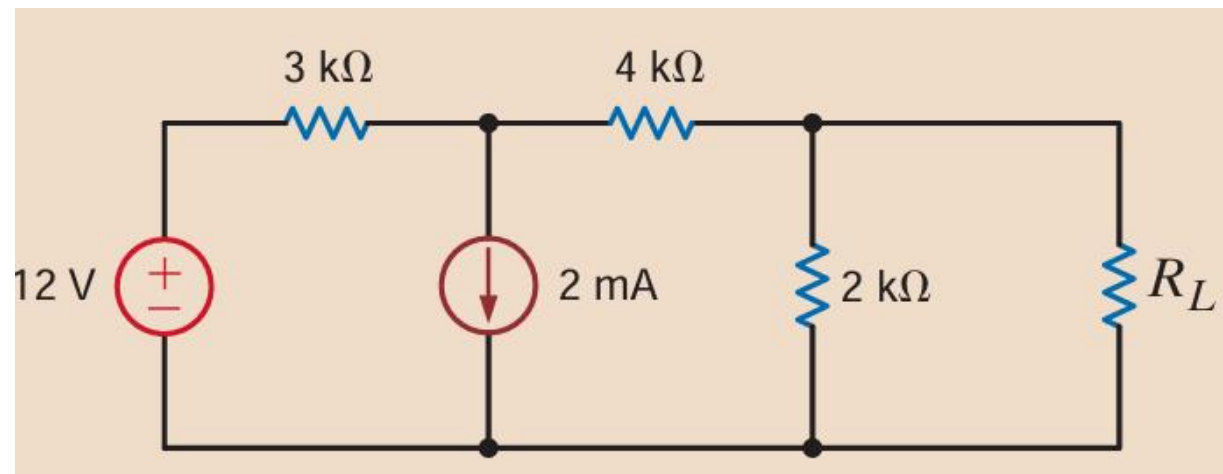
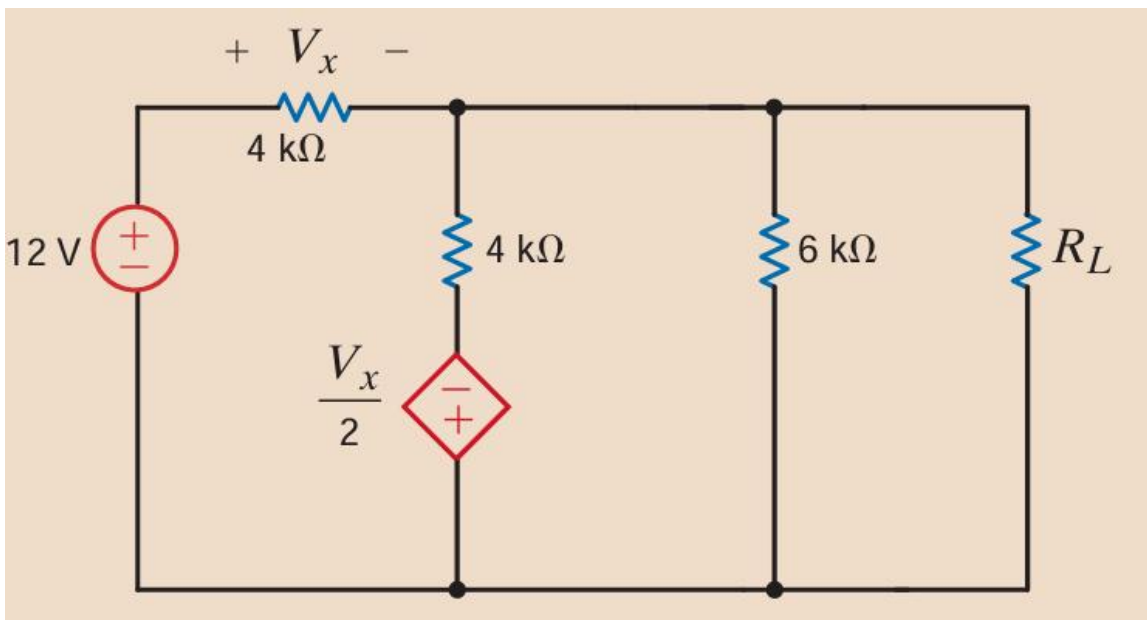
Find V_o using Thévenin's theorem



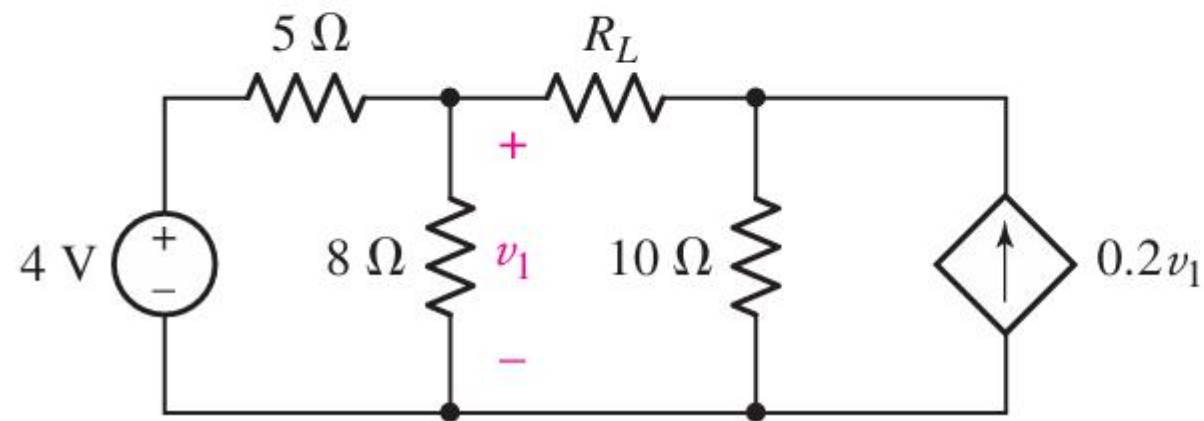
determine the Thévenin equivalent of the network in Fig. below at the terminals A-B



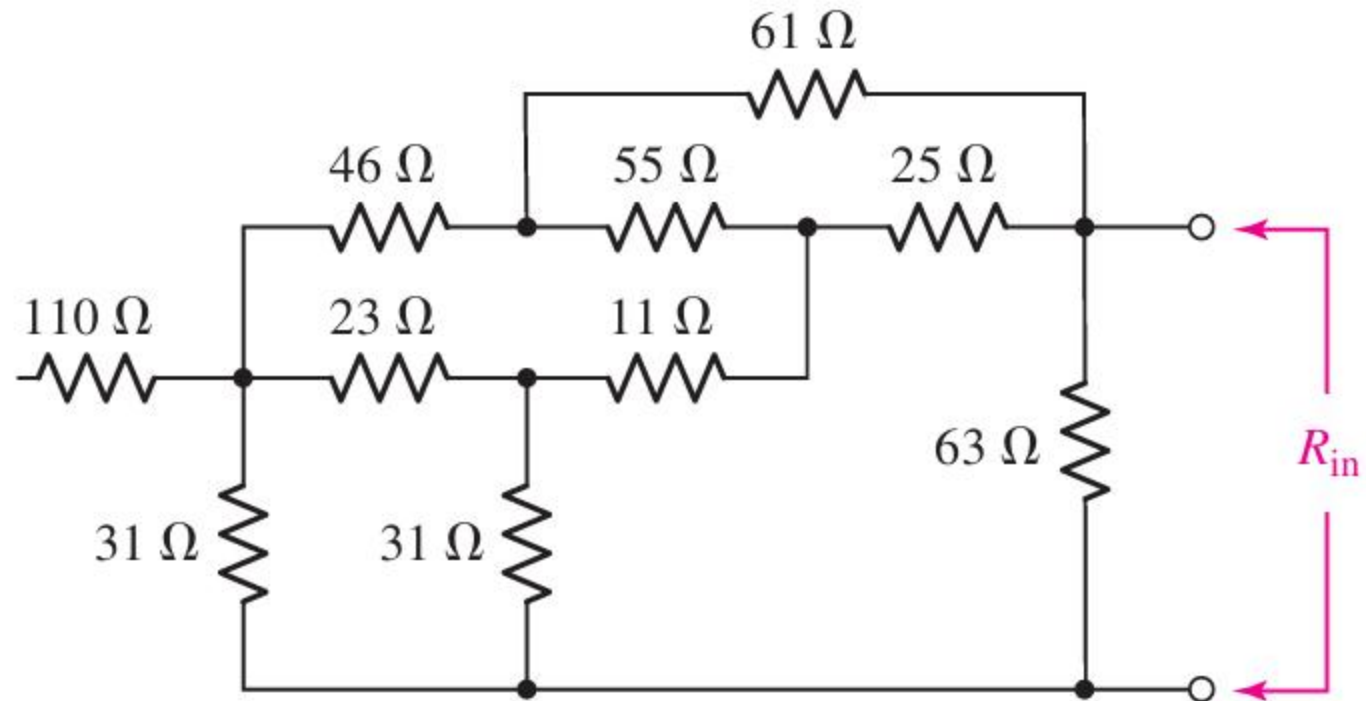
Find R_L for maximum transfer and the maximum power transferred to R_L in the circuits below.



53. Select a value for R_L in Fig. 5.93 such that it is ensured to absorb maximum power from the circuit.



61. Calculate R_{in} as indicated in Fig. 5.100.



62. Employ Δ/Y conversion techniques as appropriate to determine R_{in} as labeled in Fig. 5.101.

